If $A=[a_1 \quad a_2 \quad \ldots \quad a_n]_{1\times n}$ and $B=\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}_{n\times 1}$,then AB is a matrix of order 1×1, that gives a single element which is defined by $AB=[a_1 \quad a_2 \quad \ldots \quad a_n]\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}=[a_1b_1+a_2b_2+\ldots+a_nb_n]=\left[\sum_{k=1}^{n} a_kb_k\right]$.

For instance,

$$[1 \;\; 2 \;\; 3]\begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix}=[1(-2)+2(3)+3(5)]=[-2+6+15]=[19].$$

In general,

if $A=[a_{ij}]_{m\times n}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ and $B=[b_{ij}]_{n\times p}=\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$ then

$$AB=\begin{bmatrix} \overrightarrow{a_{11} & a_{12} & \cdots & a_{1n}} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \Bigg\downarrow \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

and the product $AB=[c_{ij}]_{m\times p}=\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$,

where $c_{ij}=[a_{i1} \quad a_{i2} \quad \ldots \quad a_{in}]\begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix}=\sum_{k=1}^{n} a_{ik}b_{kj}$, since c_{ij} is an element.

Example 7.8

If $A=\begin{bmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{bmatrix}$, compute A^2.

Solution

$$A^2=AA=\begin{bmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{bmatrix}\begin{bmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{bmatrix}=\begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

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